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GitHub repository:
https://github.com/SUKRITIRAJJ/AI-Student-Performance-and-Productivity-Analysis

Impact of AI on Student Performance & Productivity (2026)

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1. Introduction

Artificial Intelligence (AI) has become an integral part of modern education. Students increasingly rely on AI tools such as ChatGPT-4o, GitHub Copilot, Gemini Pro, Claude 3.5, and Perplexity for academic assistance, including research, coding, literature review, and exam preparation.

This project analyzes how AI adoption impacts student performance, productivity, and academic behavior using a dataset of 1,500 students. The analysis is performed using Power BI, enabling interactive visualizations and meaningful insights.

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2. Problem Statement

Artificial Intelligence (AI) is rapidly becoming a part of student life, helping with tasks such as coding, writing, and exam preparation. However, students and educators do not clearly understand how AI tools impact academic performance and productivity.

Students often question:

- Does using AI improve GPA?
- Does frequent AI usage save time?
- Which AI tools are most effective?
- Are there ethical concerns with AI usage?

Without proper analysis, these questions remain unanswered. Therefore, an interactive dashboard is required to analyze and visualize the impact of AI on student performance and behavior.

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3. Solution

To solve this problem, I developed a **two-page interactive dashboard using Power BI**, which transforms raw student data into meaningful insights.

● Page 1: Overview Dashboard

- Displays key metrics such as total students, GPA before and after AI, GPA improvement, and time saved
- Shows GPA comparison across different majors
- Visualizes AI tool usage distribution
- Highlights ethical concerns among students

● Page 2: Detailed Analysis Dashboard

- Analyzes AI usage across different tasks (exam prep, coding, etc.)
- Shows relationship between AI usage frequency and GPA
- Compares performance across different AI tools
- Provides deeper insights into student productivity

● Interactive Features

- Slicers to filter data by major, AI tool, and ethics concern
- Dynamic charts that update instantly

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4. Tech Stack

Power BI: Used for dashboard creation and visualization
DAX: Used to create calculated measures such as GPA improvement
Power Query: Used for data cleaning and transformation
GitHub: Used for project storage and sharing

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5. Dashboard Overview

The Power BI dashboard provides a comprehensive and interactive view of AI usage and its impact on student performance.

Page 1: Overview Dashboard

- KPI Cards showing key metrics such as total students, GPA before and after AI, GPA improvement, average usage, and ethics concern
- GPA comparison chart (Before vs After by Major)
- AI tool distribution (Donut Chart)
- Ethics concern breakdown (Pie Chart)
- Slicers for filtering by Major and AI Tool

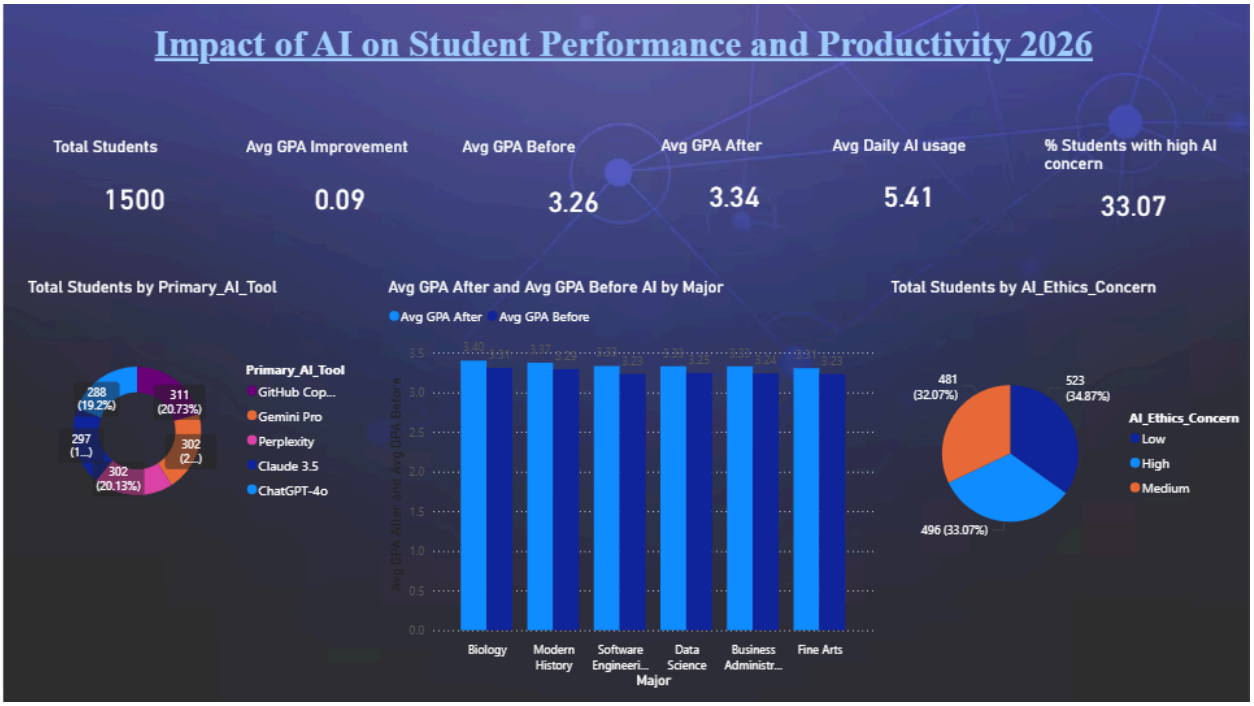

Figure 1: Overview Dashboard showing GPA trends, AI usage, and key metrics

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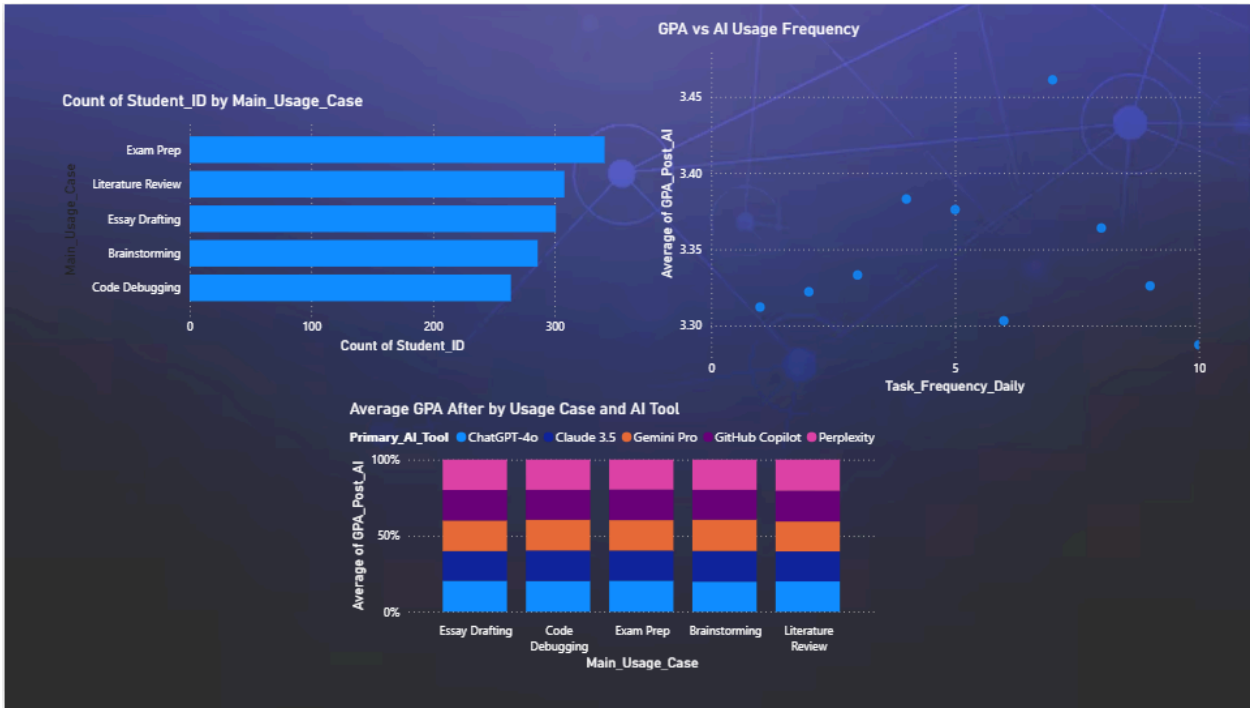
Page 2: Detailed Analysis Dashboard

- AI usage case distribution (Horizontal Bar Chart)
- Relationship between AI usage frequency and GPA (Scatter Plot)
- Average GPA after AI by usage case and tool (Stacked Column Chart)
- Additional analysis charts for deeper insights
- Slicers for filtering data



‘**Figure 2:** Detailed Analysis Dashboard showing usage patterns and performance relationships

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6. Key Insights

- Students show noticeable improvement in GPA after adopting AI tools
- Higher AI usage frequency is associated with improved academic performance
- AI tools significantly reduce time spent on academic tasks
- Students primarily use AI for code debugging, essay drafting, and exam preparation
- Different AI tools show similar effectiveness in improving performance
- A considerable percentage of students have ethical concerns regarding AI usage

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7. Unique Features of the Project

- Two-page structured dashboard for better clarity
- Analysis of both performance (GPA) and productivity (time saved)
- Use of multiple visualization types (bar, pie, scatter, stacked charts)
- Interactive filtering for better user experience
- Clean and professional dashboard design

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8. Future Improvements

- Use real-time data instead of static dataset
- Apply machine learning models to predict student performance
- Expand dataset to include more students and institutions
- Add more advanced analytics for deeper insights

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9. Conclusion

This project demonstrates the significant impact of AI on student performance and productivity. The analysis shows that AI adoption leads to improved GPA and reduced time spent on academic tasks.

While AI tools provide strong academic support, ethical concerns must be addressed to ensure responsible usage. Overall, AI has the potential to enhance learning when used effectively and responsibly.